

Part B – Weekly Capstone Assignment Template

This template is used for **each of the 14 Part B capstone submissions** throughout the semester. The structure remains the same every week so that you can focus on *thinking and judgment*, not guessing expectations.

Your goal is not to achieve the best performance, but to **reason carefully about how this week’s machine learning technique applies (or does not apply) to your project**.

1. Project Context (Brief)

Provide the same short context each week so graders can orient quickly.

- **Project Title:** Content-Based Music Recommender (Spotify Audio Features + Lyrics/Topics)
 - **Data Modality:** tabular
 - **Task Type:** recommendation (content-based similarity search)
 - **One-Sentence Goal:** Given a song, recommend other songs that sound similar using the song’s features.
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2. This Week’s Technique and Its Assumptions

- **Technique / Model Family Covered This Week:** Data preparation (train/val/test split + preprocessing) with leakage awareness
- **Key Assumptions of This Technique:** (1–2 bullets)
 - Train/val/test should simulate “future data,” so any preprocessing must be fit on train only
 - Group-based splitting by artist avoids identity leakage (no artist overlap across splits)

Fit Assessment (required):

I expect this technique to be a **good / partial / poor** fit for my project because:

This is a **good** fit because I can’t build a trustworthy recommender until the data is split safely and processed consistently. The biggest risk for my project is leakage (same artist in train and test), so the split choice matters a lot. This week’s work gives me a clean foundation for later similarity search and modeling.

3. Representation or Proxy Used

Describe how your data was represented so that this week's technique could be applied.

Examples include:

- Hand-engineered features
 - Summary statistics
 - Frozen embeddings
 - Dimensionality reduction
 - A proxy task
 - **Representation or Proxy Chosen:** A single feature vector per song (scaled numeric features + one-hot genre/topic)
 - **Why this representation was reasonable for this week:** It's simple, matches the dataset structure, and lets me do similarity-based recommendations without using identifiers like artist name or track name.
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4. What Was Attempted

Be concrete and scoped. Do not list everything you *could* have done.

- What you implemented this week I used the dataset `tcc_ceds_music.csv` and wrote a leakage-safe data prep script in `main.py`. I split the data into train/val/test **by artist** (group-based split), then did train-only preprocessing (drop ID columns, impute, scale numeric, one-hot encode genre/topic).
 - What you intentionally did *not* attempt and why I did not build the full recommender yet (top-N retrieval). This week is about data prep, not recommendation quality. I also did not use lyrics as features yet because it needs TF-IDF/embeddings and it's easy to leak information if done wrong.
 - Any constraints encountered (data, labels, compute, time) There is no user interaction data (no listens/likes), so evaluation will have to be proxy-based for now. Lyrics are huge/high-cardinality (basically unique per song), so they need a real text pipeline instead of naive encoding.
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5. Results or Observations

You may include metrics **if applicable**, but qualitative observations are also valid.

Examples:

- Evaluation metrics
 - Training behavior or convergence issues
 - Error patterns
 - Unexpected behaviors
 - The dataset had **no missing values** in the version I used, but I kept imputation code anyway as a safety step.
 - lyrics are basically **unique per song**, so I excluded them for now.
 - Group-splitting by artist makes the genre mix **slightly different** across splits (expected).
 - Basic numeric stats looked **similar across splits**, so the split doesn't seem wildly biased.
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6. Interpretation and Judgment

This section matters most.

Reflect on:

- Why the method behaved as it did
- Which assumptions held or failed
- What this reveals about your data or problem framing

This dataset seems ready for a baseline **content-based recommender** because each song already has a clean set of numeric features plus small categorical fields (**genre**, **topic**). With the artist-group split and train-only preprocessing, I can move forward without “cheating” by artist identity.

The weakest part is evaluation. Without user data (plays/likes/skips), it's hard to measure recommendation quality with standard recommender metrics. Also, leaving out lyrics is a tradeoff: it keeps things simpler and safer this week, but lyrics might matter a lot for what humans feel is “similar.”

7. Forward-Looking Adjustment

Answer **one** of the following:

- What will you keep, change, or discard before the next assignment?
- What would you try next if data or resources were not constrained? Next I will keep the leakage-safe split and preprocessing pipeline, and I'll implement the actual recommender (cosine similarity $\rightarrow$ top-N recommendations).

If I have more time, I'll add lyrics safely (TF-IDF/embeddings fit on train only) and decide on a clear proxy evaluation plan (genre/topic agreement + human spot-checking).

8. Mismatch Acknowledgment (Complete Only If Applicable)

If this week's technique was a poor fit, explain:

- Why it does not align with your project
- Evidence supporting that conclusion
- What value this attempt still provided

Not a full mismatch, but there is a real limitation: this dataset supports a content-based recommender, but it does not support standard recommender evaluation because there are no users/sessions/feedback logs. Even with that limitation, this week still mattered because I now have a clean, reproducible, leakage-safe split + preprocessing foundation to build on.

Submission Notes

- Written submission format: **Markdown or PDF**
- Code or notebooks: **optional unless explicitly requested**
- Performance is **not** graded competitively
- Clear reasoning and honest reflection matter more than results